

Quarter car suspension state space model and full state feedback control for Real-Time Processing

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Abstract— This paper presents the examination of the suspension of a quarter car model from the point of view of observability and controllability. The modeling of an active suspension with full state feedback control and the design of a state observer is planned to be used for real-time processing of the university's self-driven vehicles under development. The validation of the concept is demonstrated also. These types of mechatronics design synergy, merged with advanced engineering pedagogy are essential for professionals with appropriate qualifications from universities to enter the industry.

Keywords— quarter car model, state space model, full state feedback, state observer, real-time signal processing

I. INTRODUCTION

At the beginning of the 21st century the dominant development is the rise of AI, while according to new requirements the mathematical tools needed to solve engineering problems shifted significantly towards numerical methods [1][2][3][4]. The composition of knowledge required of engineers has changed accordingly. Some of the most important competence areas of mechatronics engineers are systems analysis, design, and control, furthermore system modelling which forms the basis for all of these areas [5][6]. We are looking for new efficient signal processing algorithms which based on stochastic modelling such as using adaptive Schur filter. New technologies are implemented to the technical progress of Industry 4.0, such as robots, autonomous transportation, and smart city concepts requires state-of-the-art algorithms for real-time signal processing of vehicles safety and reliability performance increase [7][8]. For these reasons, we examine the suspension of cars based on various aspects and design an active suspension for it. It is increasingly difficult for non-industrial parties to keep up with the car manufacturers, since these smart factories based on Industry 4.0 merge cyber technologies such as Cyber Security, manufacturing processes on the Internet of things (IoT) or Cyber-Physical Systems (CPS) [9][10][11]. There are emerging technologies in the field of physical realization of more compound and perfect, performance improvement, quality, controllability, management, and visibility of era. Quarter car system is a two degree of freedom system, where there are two bodies: Sprung (car) and Unsprung (suspension, wheel, tyre) mass. Read more details in [12][13][14].

II. VEHICLE SUSPENSION WITH AN ACTUATOR

A. Equations of motion in matrix form

First, for the state space modelling we need the equations of motion in a matrix form of the suspension model. Our model is detailed, we also take into account the elasticity of the tire and the weight of the wheel, see in Fig. 1.

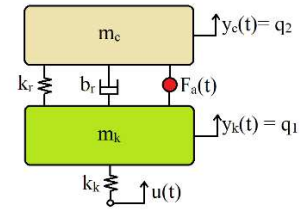


Fig. 1. Car suspension model [author's edit]

Based on Fig. 1. we can write the equations of motion using Lagrange's equations of the second kind or Newton's second law [15][8].

$$\begin{cases} I. m_k \cdot \ddot{y}_k = k_k \cdot (u - y_k) - k_r \cdot (y_k - y_c) - b_r \cdot (\dot{y}_k - \dot{y}_c) - F_a \\ II. m_c \cdot \ddot{y}_c = k_r \cdot (y_k - y_c) + b_r \cdot (\dot{y}_k - \dot{y}_c) + F_a \end{cases} \quad (1)$$

$$\begin{cases} I. \ddot{y}_k = \frac{k_k \cdot (u - y_k)}{m_k} - \frac{k_r \cdot (y_k - y_c)}{m_k} - \frac{b_r \cdot (\dot{y}_k - \dot{y}_c)}{m_k} - \frac{F_a}{m_k} \\ II. \ddot{y}_c = \frac{k_r \cdot (y_k - y_c)}{m_c} + \frac{b_r \cdot (\dot{y}_k - \dot{y}_c)}{m_c} + \frac{F_a}{m_c} \end{cases} \quad (2)$$

$$\begin{cases} I. \ddot{y}_k = \frac{k_k \cdot u - k_k \cdot y_k}{m_k} - \frac{k_r \cdot y_k - k_r \cdot y_c}{m_k} - \frac{b_r \cdot \dot{y}_k - b_r \cdot \dot{y}_c}{m_k} - \frac{F_a}{m_k} \\ II. \ddot{y}_c = \frac{k_r \cdot y_k - k_r \cdot y_c}{m_c} + \frac{b_r \cdot \dot{y}_k - b_r \cdot \dot{y}_c}{m_c} + \frac{F_a}{m_c} \end{cases} \quad (3)$$

The choice of state variables is necessary to write down the state space model.

$$x_1 = y_k, \quad \frac{d}{dt} x_1 = \dot{y}_k = x_2 \quad (4)$$

$$x_2 = \dot{y}_k, \quad \frac{d}{dt} x_2 = \ddot{y}_k \quad (5)$$

$$x_3 = y_c, \quad \frac{d}{dt} x_3 = \dot{y}_c = x_4 \quad (6)$$

$$x_4 = \dot{y}_c, \quad \frac{d}{dt} x_4 = \ddot{y}_c \quad (7)$$

B. State space representation

General state space model structure input and output equations:

$$\dot{\mathbf{x}} = \mathbf{A} \cdot \mathbf{x} + \mathbf{B} \cdot \mathbf{u} \quad (8)$$

$$\mathbf{y} = \mathbf{C} \cdot \mathbf{x} + \mathbf{D} \cdot \mathbf{u} \quad (9)$$

Using the structure above, by substituting the corresponding variables into the matrices of the model, the following equation is obtained:

$$\frac{d}{dt} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -k_k - k_r & -b_r & k_r & b_r \\ m_k & m_k & m_k & m_k \\ 0 & 0 & 0 & 1 \\ k_r & b_r & -k_r & -b_r \\ m_c & m_c & m_c & m_c \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ -1 & k_k \\ 0 & 0 \\ 1 & 0 \\ 1 & 0 \\ 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} F_a(t) \\ u(t) \end{bmatrix} \quad (10)$$

$$\mathbf{y} = \begin{bmatrix} 0 & 0 & 1 & 0 \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} + 0 \cdot u(t) \quad (11)$$

The output $\mathbf{C}$ matrix is $[0 \ 0 \ 1 \ 0]$, because we are interested in y_c output.

C. Parameters of the suspension

For this research parameters of a quarter car suspension models were used of a BMW 530i, its performance values listed in Table 1.

TABLE I. BMW 530i SUSPENSION PARAMETERS

	Front	Back
k_r [kN/m] suspension spring constant	30	31.5
b_r [Ns/m] suspension damping constant	1450	4000
k_k [kN/m] tire spring constant	340	340
m_c [kg] mass of vehicle body	408	400
m_{us} [kg] mass of wheel	48.3	45

III. OBSERVABILITY AND CONTROLLABILITY STUDY

The observability matrix:

$$\mathbf{M}_o = [\mathbf{C} \quad \mathbf{CA} \quad \mathbf{CA}^2 \quad \mathbf{CA}^3]^T \quad (12)$$

Examining the system from the point of view of observability, by numerically substituting the data in the first table into equation (12), we get:

$$\mathbf{M}_o = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 73.5 & 3.6 & -73.5 & -3.6 \\ -27486 & -45.8 & 2468.7 & 45.8 \end{bmatrix} \quad (13)$$

$$\text{rank}(\mathbf{M}_o) = 4 \quad (14)$$

So, our matrix $\mathbf{M}_o$ has a rank of 4, equal to $\mathbf{A}$'s order, thus we can say the system is observable.

The controllability matrix:

$$\mathbf{M}_c = [\mathbf{B} \quad \mathbf{AB} \quad \mathbf{A}^2\mathbf{B} \quad \mathbf{A}^3\mathbf{B}] \quad (15)$$

Examining the system from the point of view of controllability, by numerically substituting the data in the first table into equation (15), we get:

$$\mathbf{M}_c = \begin{bmatrix} 0 & -0.0207 & 0.6951 & 136.7853 \\ -0.0207 & 0.6951 & 136.7853 & -9450.7 \\ 0 & 0.0025 & -0.0823 & 1.0603 \\ 0.0025 & -0.0823 & 1.0603 & 539.5190 \end{bmatrix} \quad (16)$$

$$\text{rank}(\mathbf{M}_c) = 4 \quad (17)$$

So, our matrix $\mathbf{M}_c$ has a rank of 4, equal to $\mathbf{A}$'s order, thus we can say the system is controllable.

IV. DESIGN OF A FULL STATE FEEDBACK CONTROL

We design a state feedback controller for the original two-mass-two-spring system (see in Fig. 1.), with input $F_a(t)$ and assuming that all state variables are measurable.

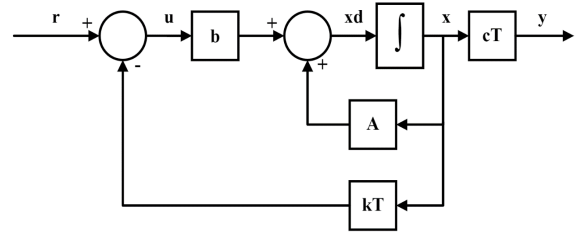


Fig. 2. Full state feedback control [author's edit]

We calculated $\mathbf{k}^T$, the feedback gain using the Ackermann-formula:

$$\mathbf{k}^T = [0 \quad 0 \quad \dots \quad 1] \cdot \mathbf{M}_c^{-1} \cdot \alpha(\mathbf{A}) \quad (18)$$

Where:

$$\alpha(\mathbf{A}) = \mathbf{A}^n + \dots + \alpha_2 \cdot \mathbf{A}^2 + \alpha_1 \cdot \mathbf{A} + \alpha_0 \cdot \mathbf{I}, \quad (19)$$

$$\mathbf{M}_c^{-1} = \frac{\text{adj}(\mathbf{M}_c)}{\det(\mathbf{M}_c)}. \quad (20)$$

The original poles of our system can be calculated from the roots of the characteristic polynomial of the matrix $\mathbf{A}$ (using the Matlab *eig()* command):

$$p_1 = -15.2734 + 85.6731i \quad (21)$$

$$p_2 = -15.2734 - 85.6731i \quad (22)$$

$$p_3 = -1.5139 + 8.1274i \quad (23)$$

$$p_4 = -1.5139 - 8.1274i \quad (24)$$

The corresponding characteristic polynomial:

$$\begin{aligned} a(s) &= \det(\mathbf{S} \cdot \mathbf{I} - \mathbf{A}) \\ &= s^4 + 33.5746 \cdot s^3 + 7733.9850 \\ &\quad \cdot s^2 + 25017.2533 \cdot s \\ &\quad + 517598.3437 \\ &= s^4 + a_3 \cdot s^3 + a_2 \cdot s^2 + a_1 \cdot s \\ &\quad + a_0 \end{aligned} \quad (25)$$

The original poles are two complex conjugate pole pairs, which cause the oscillation in the system. We replaced these with two negative real pole pairs, so that the system is set up without overshooting. Let us choose poles be the following (of course, this was preceded by a long trial and consideration):

$$p_1 = -80, p_2 = -80, p_3 = -8, p_4 = -8.$$

The radical form formed from the new poles:

$$\begin{aligned}
\alpha(s) &= (s + 80) \cdot (s + 80) \cdot (s + 8) \cdot (s + 8) \\
&= s^4 + 176 \cdot s^3 + 9024 \cdot s^2 \\
&\quad + 112640 \cdot s + 409600 \\
&= s^4 + \alpha_3 \cdot s^3 + \alpha_2 \cdot s^2 + \alpha_1 \cdot s \\
&\quad + \alpha_0
\end{aligned} \tag{26}$$

Calculation of $\alpha(\mathbf{A})$ from the ones above:

$$\begin{aligned}
\alpha(\mathbf{A}) &= \mathbf{A}^4 + \alpha_3 \cdot \mathbf{A}^3 + \alpha_2 \cdot \mathbf{A}^2 + \alpha_1 \cdot \mathbf{A} + \alpha_0 \cdot \mathbf{I} = \\
&\begin{bmatrix} 23078174.97 & -898592.29 & -2168859.32 & -16365.24 \\ 6882422944.07 & 49996387.42 & -556928533.64 & -29087071.78 \\ -3819846.34 & -1937.36 & 148756.33 & 89560.11 \\ 21426323.58 & -3443396.00 & -7788627.71 & -227694.01 \end{bmatrix}
\end{aligned} \tag{27}$$

We calculated the feedback gain from the already available data:

$$\begin{aligned}
\mathbf{k}^T &= [0 \quad 0 \quad 0 \quad 1] \cdot \mathbf{M}_c^{-1} \cdot \alpha(\mathbf{A}) = \\
&[-63048.75 \quad -6277.93 \quad -6259.58 \quad 5078.61]
\end{aligned} \tag{28}$$

We can also calculate the new $\mathbf{A}$ matrix of the controlled system:

$$\begin{aligned}
\mathbf{A}_{cl} &= \mathbf{A} - \mathbf{B} \cdot \mathbf{k}^T = \\
&\begin{bmatrix} 0 & 1 & 0 & 0 \\ -8965.81 & -160.00 & 491.52 & 135.17 \\ 0 & 0 & 0 & 1 \\ 228.06 & 18.94 & -58.19 & -16.00 \end{bmatrix}
\end{aligned} \tag{29}$$

V. SIMULINK MODEL OF THE FULL STATE FEEDBACK SYSTEM

A. Step response

After the calculation, we use Simulink to examine the response of the original and state feedback system. Our model is shown in Fig. 3.

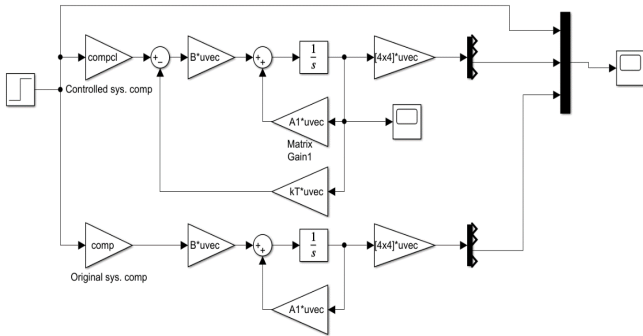


Fig. 3. Full state feedback control model [author's edit]

First, we examined the system's response to a step signal shown in Fig.4. The Step signal can be seen in blue, the unregulated, original system in orange. This will oscillate to steady state as expected from its original poles. The full state feedback-controlled system is visible in lemon yellow. It can be seen that it settles in about one-fifth as long, since its larger pole is about five times the size of its original counterpart.

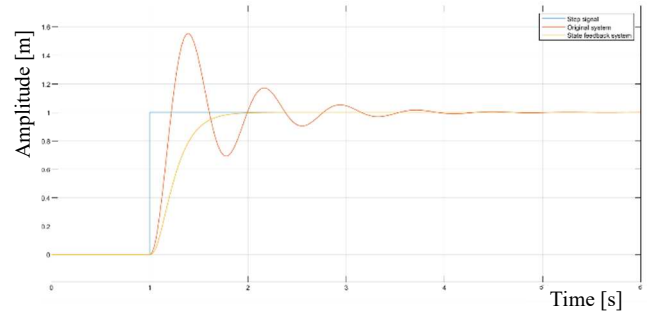


Fig. 4. Step signal responses of Simulink models [author's edit]

We also tested the status feedback with other poles, and the results obtained for this show how we can influence the setup time by changing the poles.

B. Response to speed bump model

We also test the system, which is a speed bump model, with an example that occurs very often during everyday car use. The model represents the physical model of the speed bump, with an example that occurs very often during everyday car use. The model represents the physical model of the speed bump in such a way that it gives the excitation signal based on the specified car speed and the speed bump's height, platform width, as well as the angle of ascent and descent, see in Fig. 5.

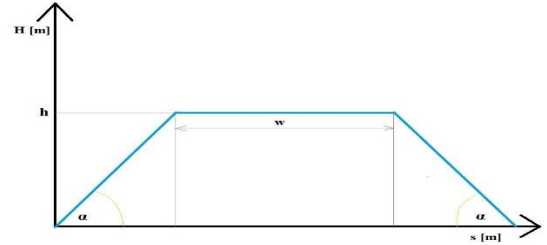


Fig. 5. Speedbump geometry [author's edit]

We also test our full state feedback model with speed bump model:

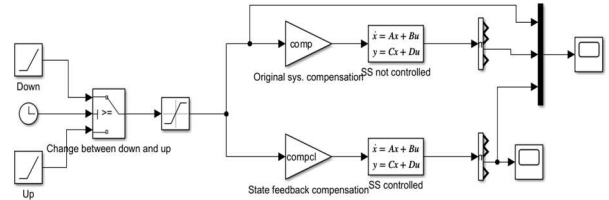


Fig. 6. Full state feedback control Simulink model with speed bump excitation [author's edit]

The used speed bump is 0.2 m high, has a 30 degree rise and fall angle, and a 2 m platform width. We made the test with 2 car speeds: 15 and 30 km/h.

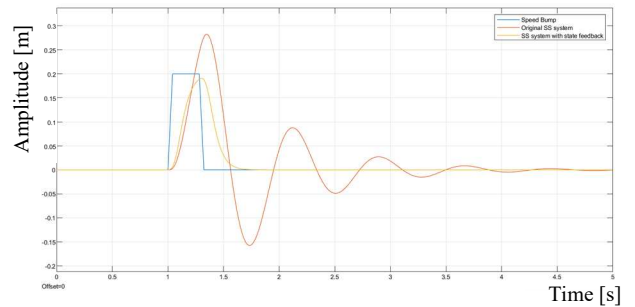


Fig. 7. Speed bump excitation response $v = 30$ km/h [author's edit]

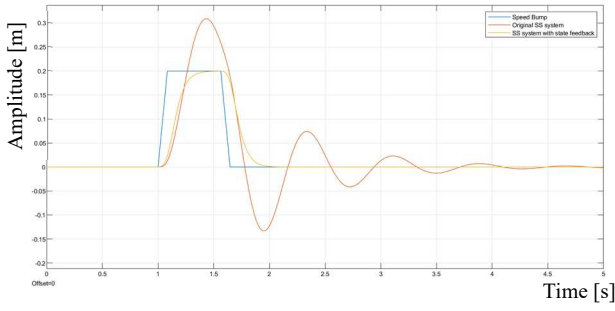


Fig. 8. Speed bump excitation response $v = 15$ km/h [author's edit]

The results obtained during the simulations are as expected. Full state feedback-controlled systems with larger poles start up faster, and also works properly with the speedbump model.

VI. DESIGN OF A STATE OBSERVER

We design a state observer for the original two-mass-two-spring system (see in Fig. 1.), with input $F_a(t)$ and an output of $y_c(t)$. Our observer model is shown in Fig. 9.

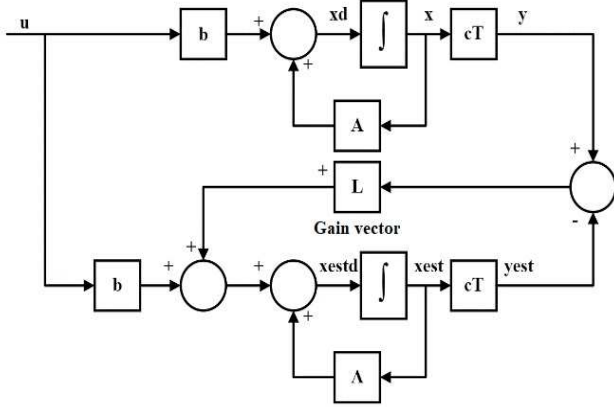


Fig. 9. State observer model [author's edit]

We defined the observability canonical forms using the duality theorem.

$$\mathbf{A}_o = \mathbf{A}_c^T \quad (30)$$

$$\mathbf{b}_o = \mathbf{c}_c \quad (31)$$

$$\mathbf{c}_o^T = \mathbf{b}_c^T \quad (32)$$

This requires the characteristic polynomial of the original system shown in (25). From that we can define $\mathcal{T}(\mathbf{a})$ auxiliary matrix:

$$\mathcal{T}(\mathbf{a}) = \begin{bmatrix} 1 & a_3 & a_2 & a_1 \\ 0 & 1 & a_3 & a_2 \\ 0 & 0 & 1 & a_3 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 33.5746 & 7733.9850 & 25017.2533 \\ 0 & 1 & 33.5746 & 7733.9850 \\ 0 & 0 & 1 & 33.5746 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (33)$$

Then the inverse of the transformation matrix $\mathbf{T}$:

$$\mathbf{T}^{-1} = \mathbf{M}_c(\mathbf{A}, \mathbf{B}) \cdot \mathcal{T}(\mathbf{a}) =$$

$$\begin{bmatrix} 0 & -0.021 & 5.28 \cdot 10^{-7} & -1.98 \cdot 10^{-5} \\ -0.021 & 5.28 \cdot 10^{-7} & -1.98 \cdot 10^{-5} & -0.0034 \\ 0 & 0.0025 & -6.25 \cdot 10^{-8} & 17.253 \\ 0.0025 & -6.25 \cdot 10^{-8} & 17.25 & -3.54 \cdot 10^{-5} \end{bmatrix} \quad (34)$$

From this, the transformation matrix $\mathbf{T}$ is:

$$\mathbf{T} = [\mathbf{T}^{-1}]^{-1} = \frac{\text{adj}[\mathbf{T}^{-1}]}{\det[\mathbf{T}^{-1}]} =$$

$$\begin{bmatrix} -0.0024 & -48.30 & -0.0096 & -5.55 \cdot 10^{-5} \\ -48.30 & 1.75 \cdot 10^{-7} & -5.55 \cdot 10^{-5} & 1.48 \cdot 10^{-6} \\ 1.75 \cdot 10^{-7} & 0.0069 & 1.48 \cdot 10^{-6} & 0.058 \\ 0.0069 & -6.33 \cdot 10^{-16} & 0.058 & 4.70 \cdot 10^{-17} \end{bmatrix} \quad (35)$$

Then we calculate the controllability canonical forms of the system's matrices:

$$\mathbf{A}_c = \mathbf{T} \cdot \mathbf{A} \cdot \mathbf{T}^{-1} =$$

$$\begin{bmatrix} -33.57 & -7733.98 & -25017.25 & -517610.66 \\ 1 & 8.33 \cdot 10^{-13} & 2.24 \cdot 10^{-13} & 0.17 \\ -2.97 \cdot 10^{-15} & 1 & -2.22 \cdot 10^{-12} & 0.00095 \\ -3.08 \cdot 10^{-16} & -1.014 \cdot 10^{-13} & 1 & -2.55 \cdot 10^{-5} \end{bmatrix} \quad (36)$$

$$\mathbf{b}_c = \mathbf{T} \cdot \mathbf{B} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} \quad (37)$$

$$\mathbf{c}_c^T = \mathbf{C}^T \cdot \mathbf{T}^{-1} = [0 \quad 0.0025 \quad 0 \quad 17.2533] \quad (38)$$

From these we determine the matrices $\mathbf{A}_o$, $\mathbf{b}_o$ and $\mathbf{c}_o^T$:

$$\mathbf{A}_o = \mathbf{A}_c^T = \begin{bmatrix} -33.57 & 1 & -2.97 \cdot 10^{-15} & -3.08 \cdot 10^{-16} \\ -7733.98 & 8.33 \cdot 10^{-13} & 1 & -1.014 \cdot 10^{-13} \\ -25017.25 & 2.24 \cdot 10^{-13} & -2.22 \cdot 10^{-12} & 1 \\ -517610.66 & 0.17 & 0.00095 & -2.55 \cdot 10^{-5} \end{bmatrix} \quad (39)$$

$$\mathbf{b}_o = \mathbf{c}_c = \begin{bmatrix} 0 \\ 0.0025 \\ 0 \\ 17.2533 \end{bmatrix} \quad (40)$$

$$\mathbf{c}_o^T = \mathbf{b}_c = [1 \quad 0 \quad 0 \quad 0] \quad (41)$$

The poles of the regulated system should be:

$$p_1 = -80, p_2 = -80, p_3 = -20, p_4 = -20.$$

The absolute value of their real part is greater than the absolute value of the real part of the poles of the original system, so they are suitable for state estimator design. The radical form formed from the new poles:

$$\begin{aligned} \alpha(s) &= (s + 80) \cdot (s + 80) \cdot (s + 20) \cdot (s + 20) \\ &= s^4 + 200 \cdot s^3 + 13200 \cdot s^2 \\ &\quad + 320000 \cdot s + 2560000 \\ &= s^4 + \alpha_3 \cdot s^3 + \alpha_2 \cdot s^2 + \alpha_1 \cdot s \\ &\quad + \alpha_0 \end{aligned} \quad (42)$$

From these we can calculate the gain vector $\mathbf{L}_o^T$, which is responsible for the feedback of the inverse system:

$$l_{0o} = \alpha_0 - a_0 = 2042401.656 \quad (43)$$

$$l_{1o} = \alpha_1 - a_1 = 294982.7467 \quad (44)$$

$$l_{2o} = \alpha_2 - a_2 = 5466.015 \quad (45)$$

$$l_{3o} = \alpha_3 - a_3 = 166.4254 \quad (46)$$

We form $\mathbf{L}_o^T$ from these:

$$\mathbf{L}_o^T = \mathbf{k}_o^T = [l_{3o} \quad l_{2o} \quad l_{1o} \quad l_{0o}] = [166.4254 \quad 5466.015 \quad 294982.7467 \quad 2042401.656] \quad (47)$$

Then $\mathbf{k}_c^T$:

$$\mathbf{k}_c^T = \mathbf{k}_o^T \cdot \mathbf{T} = [-250000 \quad -6014.34 \quad 118376.14 \quad 17097.20] \quad (48)$$

And $\mathbf{L}_c$ vector:

$$\mathbf{L}_c = \mathbf{k}_c = \begin{bmatrix} -250000 \\ -6014.34 \\ 118376.14 \\ 17097.20 \end{bmatrix} \quad (49)$$

Our Simulink model for the state observer:

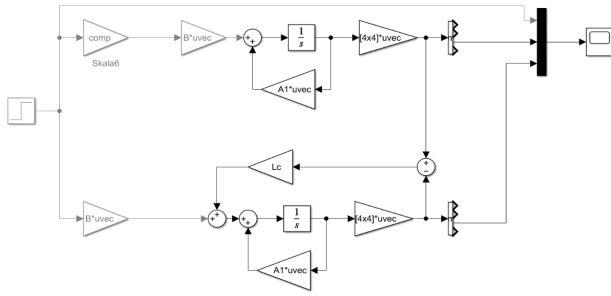


Fig. 10. State observer Simulink model [author's edit]

Then we examine the state observer with step response:

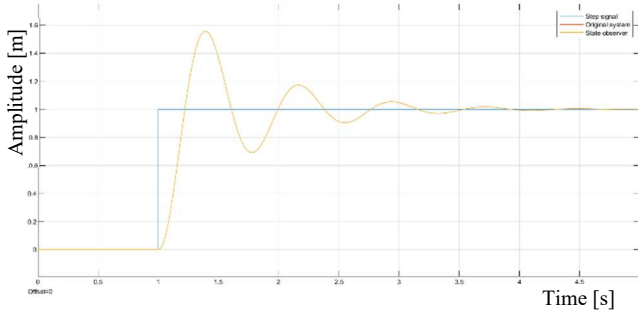


Fig. 11. State observer step response [author's edit]

The response of the original system and the state observer cannot be distinguished, they overlap.

VII. EXPERIMENTAL VALIDATION OF THE MODEL

While the simulations examined a BMW 530i vehicle, the real measurements were performed with a BMW 520d vehicle. According to the manufacturer, the two vehicles have the same mass, so this is sufficient for the tests. With the car, we investigated the response of a speedbump stimulation at different speeds. The parameters of the speedbump are known: height 0.1 m, platform width 0.2 m, elevation and descent angle 20 degrees. The simulation performed with these given values too.

The suspension of the vehicle was examined with a slow-motion camera while it was driving through the speedbump, and then the frames exported from the recording were examined in such a way that the pixel coordinates of the

position of the signal taped up on the outside of the vehicle, above the wheel, were collected and used to create the figures. Since the parameters and dimensions of the vehicle are known, we can deduce the amplitude of the oscillation, thereby correcting it compared to the simulation.



Fig. 12. Test car on the speedbump in the maximum amplitude position, 5km/h [author's photo]

The results obtained during the test and simulated are shown on a graph in Fig. 13., where the results are shown after normalization and amplitude correction.

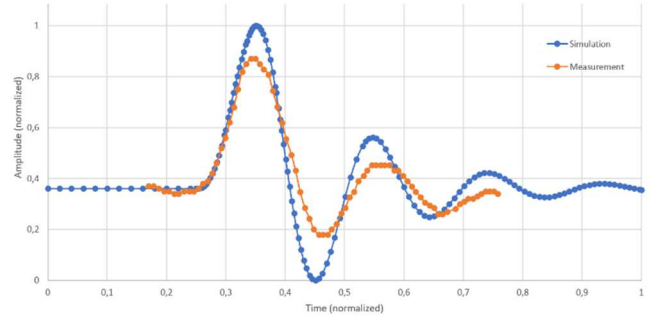


Fig. 13. Measured and simulated data of the suspension, 5km/h [author's edit]

The suspension model we used is a simplified system with no better results expected. In terms of passenger comfort, there is no significant difference between the measured and simulated results, so our model can be compared to reality. The discrepancy can also be explained by a number unknown to us, which is the aging of the shock absorber system, this may require further investigation.

VIII. SUMMARY

This paper describes the complete modeling and state feedback control of a quarter car suspension for Real-Time Signal Processing of vehicles safety and reliability performance increase. The validation of the model is proven by real measurement. This type of studies is absolute necessary for modern mechatronics education purposes of Industry 4.0 and 5.0 and for detailed analyses of the behaviour and automatic control systems development of mobile system such as mobile radar platforms. During the design, it was demonstrated not only control technology, but also electronic and mechanical problems, taking into account the mechatronic synergy. The acquisition of these skills is essential in order for professionals with appropriate qualifications from universities to enter the industry.

The efficiency of the algorithms could be checked by alternative velocity measurement and traffic monitoring systems such as introduced in [16]. This task is also part of a

work to be submitted, which is an integral part of the engineering self-driven car development activities are covering the mechatronics engineer training at the University of Debrecen.

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